

Referee Report: The Saving Glut of the Rich

Review of Mian, Straub, and Sufi (July 2025)

Recommendation: revise and resubmit. This is an ambitious, original, and highly relevant paper. Its central innovation is to combine Financial Accounts data with a Leontief-style operator that traces claims through intermediary cross-holdings to their ultimate owners and uses. This reveals a major within-household shift that aggregate national accounts conceal: the rise in top-1-percent saving and its counterpart in household and government debt. The treatment of valuation changes, retained corporate earnings, and national-accounts closure is particularly valuable. Our reconstruction using the older supplied inputs reproduces the main patterns closely, increasing confidence that the paper captures a real and economically important fact. I recommend publication after a focused revision addressing the following measurement and interpretation issues.

1. Show saving rates over time and by inflation-adjusted wealth levels

Figure 6 compares two long-period averages by wealth percentile. This is a useful summary, but percentile rank is not the only relevant horizontal axis. If households become richer over time and consumption rises less than proportionally with wealth, an increasing saving rate at high ranks could partly reflect movement along a stable, non-homothetic saving schedule rather than a change in behavior conditional on resources. The paper should therefore show the same saving-rate definition against inflation-adjusted wealth levels, ideally using fixed real-wealth bins in addition to percentile bins.

Our diagnostic re-plots the same percentile-cohort saving rates at their mean net wealth per adult in 2018 dollars (Referee Figure 1). The result does **not** make the post-1982 divergence disappear. The pre-1982 90th-percentile cohort has mean wealth of about \$303,000 and a 31.5 percent saving rate, while the post-1982 80th-percentile cohort has similar mean wealth of about \$286,000 but saves only 7.4 percent. The top 1 percent moves from about \$3.0 million to \$6.9 million in mean wealth while its saving rate rises from 28.8 to 36.2 percent. These comparisons suggest both a rightward shift in wealth and a change in the saving-rate profile. They remain descriptive because cohort means are not fixed-dollar bins and the data are repeated cross-sections; this is precisely why the authors should perform the fixed-real-wealth-bin exercise on their exact 1963–2019 microdata.

The two-period average also hides considerable time variation (Referee Figure 2). In our older-input reconstruction, the five-year top-1 rate is 35.6 percent in the window ending in 1982, peaks at 68.5 percent in the window ending in 2008, and falls to 24.1 percent by 2016. The bottom-40 rate moves from -3.3 to -15.3 and back to -3.2 percent over the same windows. This is consistent with strong crisis-era movements at both ends, in opposite directions, but it is not a causal Great Recession estimate: overlapping windows blur timing and the 2008 window contains four pre-crisis years. The paper should add annual or non-overlapping windows, recession shading, and a decomposition into financial assets, housing, debt write-downs, and the equity-valuation residual.

A standard logarithmic color scale is invalid because saving rates can be zero or negative. I suggest a linear, zero-centered main heatmap and a clearly labelled symmetric-logarithmic or inverse-hyperbolic-sine diagnostic, accompanied by selected-percentile lines. If the intended object is *change*, the heatmap should plot deviations from each percentile’s pre-1982 mean rather than levels.

2. Bound what the unveiling and saving assumptions identify

The proportional allocation rule is transparent, but claims issued by an intermediary differ in seniority, maturity, collateral, guarantees, and risk. The operator therefore identifies proportional ultimate claim ownership, not necessarily loss incidence, control, or causation. The paper should report the observed, balanced, and imputed shares of the direct matrix; the residual share; and the sensitivity of Figures 8–10 to perturbations of the largest imputed links and alternative instrument-specific allocations.

Likewise, NIPA closure is an essential accounting check but does not identify how the residual equity valuation term is distributed. Because equity ownership is concentrated, plausible alternatives for equity inflation, corporate saving, housing returns, and debt write-downs should be propagated into uncertainty or sensitivity bands for Figures 5 and 6. These diagnostics are especially important around 2008. Finally, causal verbs such as “fueled” or “drove” should be reserved for identified mechanisms; the baseline establishes that saving is ultimately *absorbed by* or *held against* particular debt claims.

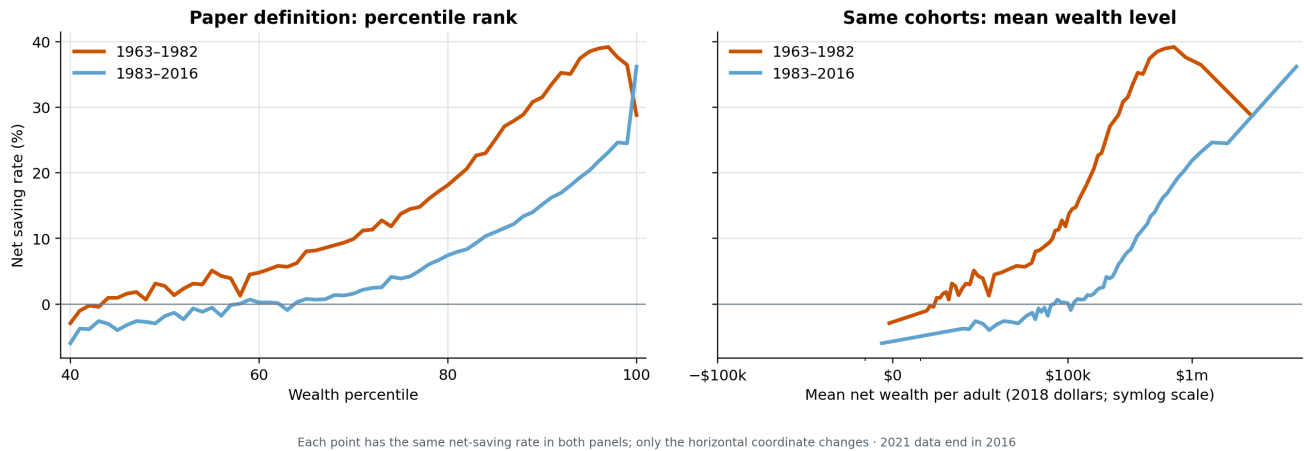
3. Provide a version-matched replication package

The February 2021 package reproduces its own paper version, but it generally ends in 2016 and uses the earlier round-by-round unveiling. The revised paper should archive the exact 2025 inputs, completed annual matrices, crosswalks, valuation construction, and figure scripts behind the 2019 results. This is particularly important for a paper whose methodology is itself a major contribution.

Overall, these requests would sharpen rather than overturn the main finding. The paper offers a useful framework and compelling evidence that rising inequality has changed the composition and ultimate use of saving.

Referee Figure 1 — Saving rates by rank and real wealth

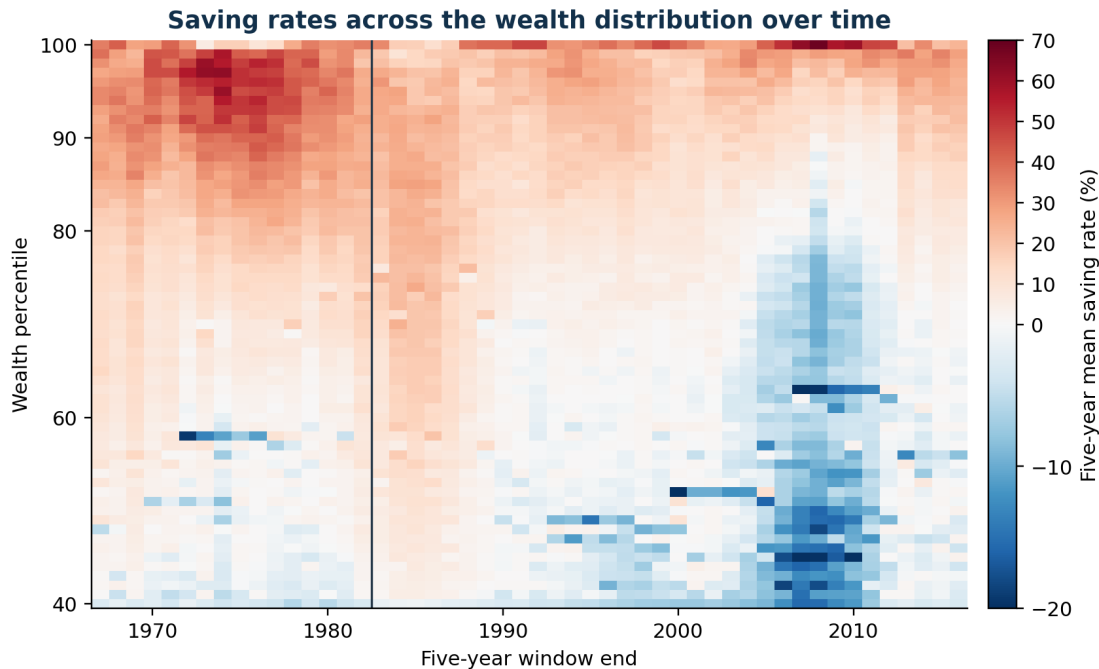
Figure 6: the denominator and numerator are held fixed



Finding. Re-indexing the same cohorts by mean real wealth does not remove the post-1982 divergence. The comparison is descriptive and does not replace the proposed fixed-real-wealth-bin estimate.

Source: generated by the independent Figure 6 reconstruction from February 2021 input-vintage data through 2016. Net wealth is measured per adult in 2018 dollars.

Referee Figure 2 — Saving rates over time



Finding. Five-year saving rates vary substantially over time and across the wealth distribution, including sharp crisis-era movements near both tails. Overlapping windows do not identify a causal recession effect.

Source: generated by the independent Figure 6 reconstruction from February 2021 input-vintage data through 2016. Colors show trailing five-year mean net saving rates.

Replicating the Saving Glut of the Rich

Independent reconstruction of selected results in Mian, Straub, and Sufi (July 2025)

Method and data

The empirical problem is to move from observed holders to *ultimate* owners. For direct ownership matrix $\bar{M}_t$ —rows are issuers and columns are holders—the revised paper sums every intermediary path. With Q_t the intermediary block and B_t the direct exits to terminal owners, our Python implementation solves

$$\Omega_t = (I - Q_t)^{-1} B_t, \quad \rho(Q_t) < 1.$$

Financial Accounts/FWTW matrices identify bilateral stocks; DINA allocates the household sector across wealth groups; NIPA closes aggregate income and saving; and asset returns separate active saving from valuation changes, $\Theta_{g,t} = \sum_a (W_{g,t}^a - W_{g,t-1}^a - \pi_t^a W_{g,t-1}^a)$.

We first ran the complete February 2021 Stata master under StataNow SE 19.5. After portability-only path changes, dependency bootstrapping, and suppressing one interactive browser call, it completed and regenerated the major outputs for the paper version it accompanies. That package generally ends in 2016 and uses seven stage-specific unveiling rounds. The 2025 paper instead reports 34 instruments, 23 intermediaries plus four ultimate owners, a revised completion, a full matrix/Leontief solve, and data through 2019. We therefore reimplemented the 2025 definitions in Python from the earliest usable supplied inputs and label every remaining version boundary.

Replication results

Exhibit	Evidence layer	Result	Binding limitation
Figure 1	Old-input reconstruction	Close level match: $r = 0.995$ and MAE = 5.3 pp over 1963–2016.	Exact 2025 completed matrices and indirect-share denominator are unavailable.
Figure 5	Old-input reconstruction	High fidelity: top 1% / bottom 99% correlations 0.992 / 0.997; MAE 0.40 / 0.42 pp.	Same saving transformation, but the authors’ supplied inputs stop in 2016.
Figure 8	Same-vintage method reconstruction	Trend and borrowing match: $r = 0.994/0.995$ and MAE = 2.33/1.31 pp for top 1% / bottom 99%.	The kit saves 351 direct fields, not the full 34-instrument by 27-sector cube; coarsening, signed-cell clipping, and an unassigned owner remain.

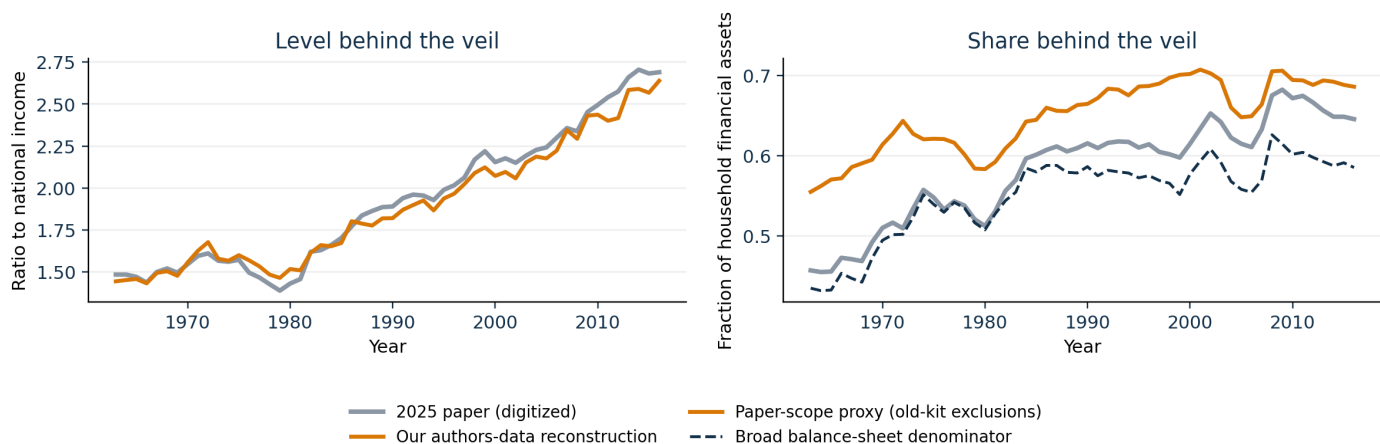
Interpretation and reproducibility boundary

These are version-bounded reconstructions, not exact numerical reproductions. Figures 1 and 5 recover the main 2025 patterns from older author-supplied inputs. For Figure 8, restarting from saved direct cells and applying the full inverse yields 2007 top-1 lending of 11.47 pp, versus 18.02 pp in the paper, while bottom-99 debt is -41.03 pp versus -38.76 pp. Holding the 2021 vintage as fixed as the saved fields permit, replacing seven rounds with the full inverse changes the two series by only 0.26 and 0.75 pp on average. The remaining gap is therefore mainly a data-vintage, completion, taxonomy, and allocation boundary rather than finite-round truncation. A current-public-FWTW rerun is a separate robustness exercise, not an exact recovery of the authors’ values.

All empirical series, comparisons, and displayed metrics are generated by code. The principal entry points are `build_figure1_authors_data.py`, `build_figure5_authors_data.py`, and `build_figure8_2021_direct.py`; tests cover matrix conservation, group partitions, valuation/saving closure, normalization, and debt signs.

Figure 1 — Indirect household ownership

Figure 1: authors-data level match, denominator remains version-specific

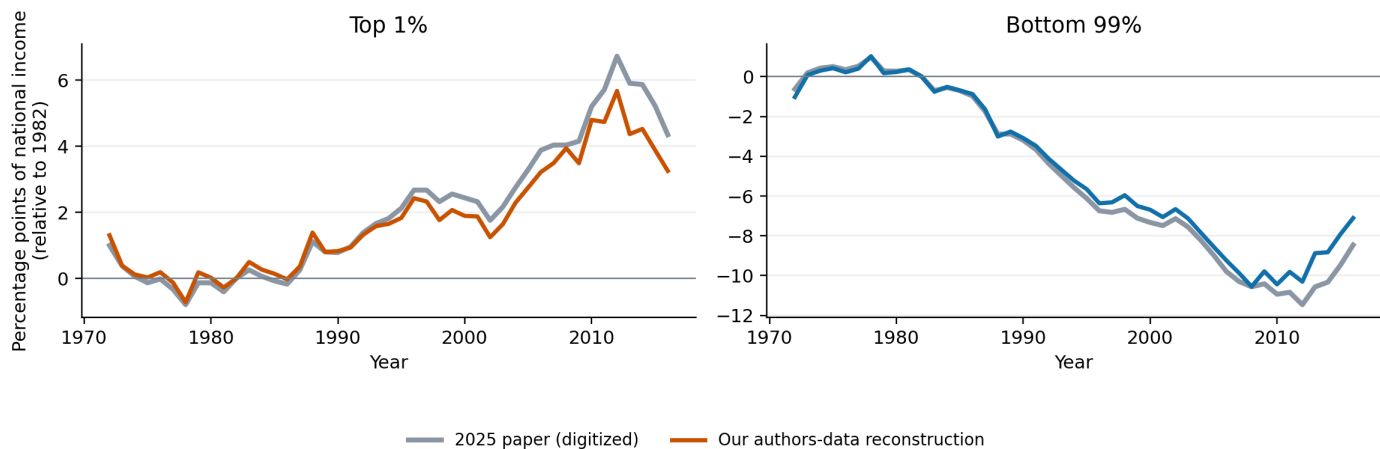


Finding. The old-input stock reconstruction follows the revised Figure 1 level closely. The plotted share remains an explicitly labelled old-kit denominator proxy.

Source: generated by `Code/scripts/build_figure1_authors_data.py`; paper panel digitized from the July 2025 PDF.

Figure 5 — Saving by wealth group

Figure 5: same transformation, older data stop in 2016

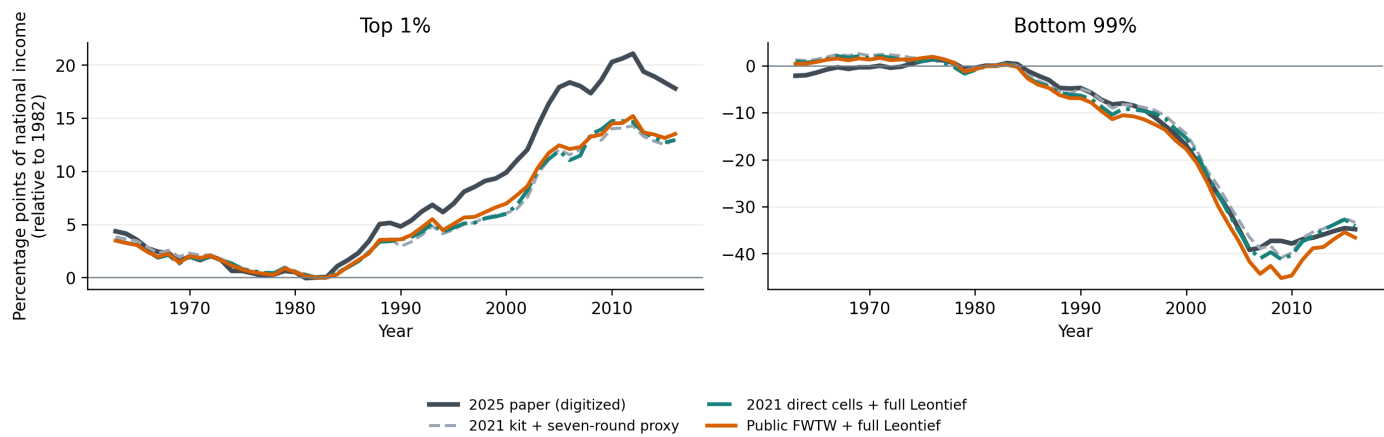


Finding. The 2025 saving transformation applied to 2021-kit inputs reproduces the divergence between top-1 and bottom-99 saving through 2016.

Source: generated by `Code/scripts/build_figure5_authors_data.py`; paper curves digitized from the July 2025 PDF.

Figure 8 — Net household debt by wealth group

Figure 8: separating the operator experiment from the public-data rerun



Finding. All three routes reproduce the post-1982 split. The primary same-vintage route restarts before unveiling and applies the full inverse; the seven-round result is its lineage benchmark and the public-FWTW route is a mixed-vintage robustness check. The small same-vintage operator gap shows that missing 2025 matrix coverage, rather than truncation alone, drives the remaining top-1 level difference.

Source: generated by Code/scripts/build_figure8_method_comparison.py; paper curves digitized from the July 2025 PDF.

From Unveiled Ownership to Unveiled Risk

Did financial intermediation transform rich saving into middle-class mortgage debt?

Motivation and research question

Mian, Straub, and Sufi (MSS) document three connected changes. Figure 1 shows the indirectly held share of household financial assets rising from about 45% in 1960 to 70% in 2007. Figure 6 shows the top 1% saving rate rising from 43% to 54% after 1982 while the rest falls from 14% to 7%. Figure 8 unveils a widening top-lender/middle-borrower position before 2008. Yet the ownership operator is payoff-blind: it traces who owns a dollar claim, not whether intermediation reduced its risk, relaxed credit supply, weakened screening, or shifted losses through capital and guarantees.

Question. Did pooling, securitization, and secondary-market liquidity convert the saving glut of the rich into cheaper and more abundant mortgages, thereby lowering middle-wealth households' saving rates; and who ultimately bore the resulting state-contingent losses?

A borrower–saver–intermediary model

Let mortgage i pay $R_{it} = \bar{R}_t - \eta_t - \varepsilon_{it}$, with aggregate risk $\sigma_{A,t}^2$ and idiosyncratic risk $\sigma_{I,t}^2$. Financialization ϕ_t permits a pool of $N(\phi_t)$ loans, so

$$\text{Var}(\bar{R}_{N,t}) = \sigma_{A,t}^2 + \frac{\sigma_{I,t}^2}{N(\phi_t)}, \quad r_t^L - r_t^f = c(\phi_t) + \gamma_t \text{Var}(\bar{R}_{N,t}) + \mu(\phi_t, e_t, k_t, g_t).$$

Pooling removes idiosyncratic risk, not the aggregate housing shock. The loan spread also contains funding cost c , screening e , loss-absorbing capital k , guarantees g , and an agency/opacity wedge μ . Hence $\partial(r^L - r^f)/\partial\phi$ is ambiguous: intermediation can reduce funding and diversification costs while weakening screening or reallocating capital.

A middle household chooses consumption, housing, and mortgage debt subject to $b \leq \lambda(\phi_t, e_t)q_t h$. Its debt choice maps directly into Figure 6:

$$s_{gt}^{L,m} = - \frac{L_{gt}^m - (1 - WD_{gt}^m)L_{g,t-1}^m}{Z_{gt}^d},$$

where WD_{gt}^m is the write-down rate and Z_{gt}^d is group disposable income including attributable corporate income. Rich savers choose direct and intermediated claims; market clearing links their saving to mortgage funding, but ownership alone is not a causal loan-supply equation.

The methodological increment: a separate loss operator

MSS compute nominal ultimate ownership as $\Omega_t = (I - Q_t)^{-1}H_t$. For state ω , instead let $P_t(\omega)$ pass marginal losses between intermediaries and $C_t(\omega)$ pass them to terminal owners after capital, seniority, and guarantees. For primary losses $\ell_t^0(\omega)$,

$$\ell_t^O(\omega) = \ell_t^0(\omega)(I - P_t(\omega))^{-1}C_t(\omega), \quad P_t \mathbf{1} + C_t \mathbf{1} + a_t = \mathbf{1},$$

where a_t is intermediary-buffer absorption. This local, piecewise-linear map keeps three objects distinct: face-value ownership, ex ante expected risk, and realized loss incidence. Large shocks require an explicit nonlinear waterfall as buffers are exhausted.

Relation and novelty

The generic links from securitization to mortgage supply and from inequality to household debt are well studied. The candidate contribution is narrower: *jointly discipline distributional saving, ultimate claim ownership, and wealth-group loss incidence, then quantify the intermediation counterfactual*. It opens the intermediation block in *Indebted Demand* and answers MSS's call to study why excess saving financed leverage rather than investment.

Mortgages Select the Pre-2008 Laboratory

February 2021 inputs, full Leontief unveiling, 1963–2016

The project separates home mortgages from broad consumer credit for four wealth groups and applies the paper’s write-down factors. Selected estimates below are generated, not hand-entered; all values are percentage points of aggregate national income.

Object	Debt category	Top 1%	Next 40%	Bottom 50%
Liability-stock change, 1982–2007	Mortgage	+4.50	+23.63	+13.85
	Consumer credit	0.00	+2.92	+4.04
Annual active-saving contribution, 1998–2007	Mortgage	−0.43	−3.09	−2.28
Unveiled net-position change, 1982–2007	Mortgage	+8.69	−19.02	−13.73

Interpretation. Mortgages dominate the boom-era debt build-up, the debt-related saving drag, and the widening top-lender/middle-borrower split. For the bottom 99%, the mortgage saving contribution then collapses from −6.36 in 1998–2007 to −0.16 in 2008–2016, so the model should target the pre-crisis regime. This is accounting incidence, not a causal financialization estimate.

Empirical and quantitative sequence

- 1. Close the MSS measurement identity.** Use the exact Figure 6 group-income denominator and percentile bins; require mortgage plus consumer credit to recover total debt saving and all asset classes to recover the complete saving rate.
- 2. Identify the intermediation channel.** Measure securitization, loan sales, GSE/private-label shares, originator funding and capital, retention, and guarantees separately. Estimate approval, rate, amount, and performance responses using conforming-loan eligibility or predetermined lender exposure. Test cutoff manipulation, borrower sorting, and the lender-exposure exclusion restriction.
- 3. Estimate the model and unveil losses.** Match Figures 1, 6, and 8; mortgage spreads and originations; intermediary funding/capital; and default/loss severity. Simulate nominal ownership and state-contingent losses, enforcing loss conservation across security holders, intermediary capital, government guarantees, and the rest of the world.
- 4. Run the counterfactual.** Holding 2007 incomes, collateral, and borrower fundamentals fixed, replace the 2007 intermediation/risk-bearing structure with its 1982 counterpart. Measure the implied change in middle mortgage debt and Figure 6 saving, then separate intermediation from inequality, collateral, and agency/guarantee wedges.

Required data

Source	Object	Principal limitation
MSS DINA, Financial Accounts/FWTW, DFA	Saving by wealth, ownership network, mortgage stocks and write-downs	Nominal claims do not reveal loss waterfalls
HMDA + Call Reports	Applications, approvals, lender funding, capital, charge-offs	HMDA lacks wealth; nonbank exposure is incomplete
FHFA/GSE limits and loan-performance files	Eligibility, securitization, default, recovery, loss severity	GSE loans do not cover private-label securitization
SCF/DFA + local house prices	Borrower wealth rank, portfolios, collateral and housing risk	Triennial survey; house prices are endogenous

Private-label pool/tranche and servicing data would sharpen risk incidence but may require commercial access. The public minimum combines the existing MSS pipeline with HMDA, Call Reports, conforming limits, GSE performance, SCF/DFA, and house prices.

Evidentiary boundary

The proposal does not assume that financialization caused top saving or that more credit reduced welfare. Pooling can lower idiosyncratic risk while leaving aggregate risk intact; securitization may relax funding constraints while weakening screening. The contribution is to estimate these offsetting forces and distinguish who owns the claim from who bears the loss.